

# A BEALE–KATO–MAJDA CRITERION FOR RIGHT-INVARIANT $H^s$ METRICS ON $\text{SDiff}(\mathbb{T}^3)$

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ABSTRACT. Fix the bracket of  $\text{SDiff}(\mathbb{T}^3)$  and let the inertia operator range over the isotropic Fourier multipliers  $a(|k|) = |k|^{2s}$ , so that  $s = 0$  is the incompressible Euler equation. We prove a continuation criterion for  $s > 0$ : with  $m$  the momentum and  $w = \text{curl } m$ , a solution persists as long as  $\int_0^T \|w\|_{L^p} dt < \infty$  for one exponent  $p > 3/(2s)$ . Two ingredients. The first is a reduction:  $w$  obeys the *unmodified* vortex-stretching equation for every  $s$ , so that all the  $s$ -dependence sits in the recovery of the velocity from  $w$ , a Fourier multiplier of order  $-1 - 2s$ . The second is that for  $s > 0$  the resulting generalised Biot–Savart law is smoothing, its kernel locally integrable, and  $\|\nabla u\|_{L^\infty} \lesssim \|w\|_{L^p}$  with no logarithm. The classical Beale–Kato–Majda criterion is the endpoint  $s \rightarrow 0^+$ , where the admissible  $p$  escapes to infinity and the logarithm of Brezis–Gallouët–Wainger becomes unavoidable. Tao observed that no such criterion was available for generalised Euler equations and did not attempt one. We also record what the criterion does not do: the conserved energy is  $\|w\|_{\dot{H}^{-1-s}}^2$ , which dominates  $\|\nabla u\|_{L^\infty}$  exactly when  $s \geq 5/2$ , recovering the geodesic completeness threshold of Bruveris–Vialard from the analytic side and confirming that the criterion alone moves nothing.

## 1. INTRODUCTION

Arnold’s theorem realises the incompressible Euler equation as the geodesic flow of the right-invariant  $L^2$  metric on the group of volume-preserving diffeomorphisms [1], in the rigorous form given by Ebin and Marsden [11]. Fixing that group and varying the metric gives a one-parameter family

$$a(|k|) = |k|^{2s}, \quad s \in \mathbb{R}, \tag{1}$$

of right-invariant metrics on  $\text{SDiff}(\mathbb{T}^3)$ , with Euler at  $s = 0$ . The family is the three-dimensional, volume-preserving analogue of the  $H^s$  metrics on  $\text{Diff}(S^1)$ , where the behaviour is understood at both ends: shocks at  $s = 0$ , wave breaking at  $s = 1$ , and geodesic completeness for  $s \geq 3/2$  [12, 2]. In three dimensions only the smoothing end is settled. For  $s > \frac{1}{2} \dim M + 1$ , that is  $s > 5/2$ , the geodesic flow of a strong right-invariant  $H^s$  metric is complete, on the volume-preserving subgroup as well as on the full group [8, Rems. 7.3 and 7.5, Cor. 7.6]; below  $5/2$  nothing is known in either direction, and  $s = 0$  is the Euler problem itself. How far that threshold can be displaced is asked in [15, Question 7.2], which this note was written to make approachable and does not answer.

Any attempt to move that threshold has to begin with a continuation criterion: a statement that a solution persists for as long as some quantity remains integrable in time. For  $s = 0$  this is the theorem of Beale, Kato and Majda [3], which replaces  $\|\nabla u\|_{L^\infty}$  by  $\|\omega\|_{L^\infty}$  at the cost of a logarithm. For the family (1) no such criterion appears to have been written. Tao, considering the wider class of generalised Euler equations in which the vector potential operator is an arbitrary operator of order  $-2$ , records the gap explicitly: analogues of at least some of the known refinements should be establishable, the generalised Biot–Savart law obeying estimates

very similar to the true one, but he does not attempt it [17, Rem. 1.10]. This note supplies the criterion for (1) with  $s > 0$ .

**What is proved.** Four things.

*The reduction* (Proposition 2.2). The field  $w := \operatorname{curl} m$ ,  $m$  the momentum, satisfies

$$\partial_t w + (u \cdot \nabla)w = (w \cdot \nabla)u, \quad \operatorname{div} w = 0,$$

the vortex-stretching equation of the *unmodified* Euler equation, for every  $s$ . It is an identity of vector calculus valid for any  $m$  whatever and any divergence-free  $u$ ; it does not use  $m = Au$ . So the whole  $s$ -dependence is carried by the constitutive law  $w \mapsto u$ , which by Lemma 2.4 is a Fourier multiplier of order  $-1 - 2s$ , whence  $\nabla u = T_s w$  with  $T_s$  of order  $-2s$ .

*The kernel estimate* (Lemma 3.1). For  $s > 0$  the operator  $T_s$  is smoothing, its convolution kernel is bounded by  $|x|^{2s-3}$  and hence lies in  $L^{p'}$  exactly when  $p > 3/(2s)$ , so that  $\|\nabla u\|_{L^\infty} \lesssim_{s,p} \|w\|_{L^p}$  with no logarithm.

*The a priori estimate* (Lemma 4.1, Proposition 4.3 and Theorem 6.2). For every  $s > 0$  and every  $r > \frac{5}{2}$ , with no threshold in between,

$$\frac{d}{dt} \|w\|_{H^r} \lesssim (\|\nabla u\|_{L^\infty} + \|w\|_{L^\infty}) \|w\|_{H^r},$$

both terms being controlled by  $\|\nabla u\|_{L^\infty}$  alone, so that finiteness of  $\int_0^T \|\nabla u\|_{L^\infty}$  forbids breakdown at  $T$ .

*The local theory* (Section 5). Existence, uniqueness and the blowup alternative, none of them assumed. Combining the four gives the criterion, Theorem 6.3.

The whole degeneracy sits in  $p$ . The admissible exponent  $p > 3/(2s)$  escapes to  $\infty$  as  $s \rightarrow 0^+$ , and at  $s = 0$  no finite  $p$  repairs the estimate: one must buy it back with the logarithm of Brezis–Gallouët–Wainger, so that the classical criterion is the endpoint of this family rather than an instance of it. The a priori estimate, by contrast, is uniform in  $s \geq 0$  and needs only  $r > \frac{5}{2}$ .

**What is not proved, and what this does not do.** Nothing about the local theory is assumed. Existence is proved by Galerkin truncation (Theorem 5.3), uniqueness by an  $L^2$  argument (Proposition 4.5), and the blowup alternative follows (Proposition 6.1). Solutions are taken in the class  $\mathcal{X}_T^r$  of Definition 5.1, bounded in  $H^r$  and continuous in every  $H^{r'}$  with  $r' < r$ , which is what the construction delivers and what every estimate here uses; such solutions lie in  $H^r$  at every time and are weakly continuous into it (Lemma 5.2). They are also strongly continuous into  $H^r$ , by the argument of Bona and Smith, which we omit as unused rather than unavailable. The commutator and product estimates of Kato–Ponce and Moser type are quoted, not proved, as are the Bernstein inequalities and the kernel asymptotics of [16]. Section 7 records the negative half: the conserved energy is  $\|w\|_{\dot{H}^{-1-s}}^2$  and a two-line count shows that it dominates  $\|\nabla u\|_{L^\infty}$  precisely when  $s \geq 5/2$ , so the criterion recovers the completeness threshold of [8] and moves it nowhere. That is the outcome to insist on: a criterion improving the threshold by itself, continued to  $s = 0$ , would prove global regularity for three-dimensional Euler.

**Verification.** The identities of Section 2 are checked spectrally on a fixed  $24^3$  grid with band-limited data, to relative errors between  $10^{-16}$  and  $10^{-18}$ , by `bkm_check.py`. The kernel claim underlying Lemma 3.1 is checked separately by `kernel_check.py`; see Remark 3.2. The remaining estimates, in Sections 4–6, are not numerical.

## 2. SETUP AND THE REDUCTION

Let  $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$  and let  $\mathfrak{g}$  be the space of smooth, divergence-free, mean-zero vector fields on  $\mathbb{T}^3$ , with the Jacobi–Lie bracket  $[u, v] = (u \cdot \nabla)v - (v \cdot \nabla)u$  and the  $L^2$  pairing. For  $s \in \mathbb{R}$  let  $A = A_s$  be the Fourier multiplier with symbol  $a(|k|) = |k|^{2s}$ , symmetric and positive definite on  $\mathfrak{g}$  because the mean mode is excluded. Write  $m = Au$  and let  $P$  be the Leray projection. All fields below are mean-zero, so that  $\|\cdot\|_{H^\sigma}$  and  $\|\cdot\|_{\dot{H}^\sigma}$  are equivalent, and we write  $\Lambda = (1 - \Delta)^{1/2}$ . The Euler–Arnold equation of the metric  $\langle A\cdot, \cdot \rangle$  is

$$\dot{m} = \text{ad}_u^* m, \quad u = A^{-1}m. \quad (2)$$

**Lemma 2.1.** *For  $u \in \mathfrak{g}$  and any vector field  $m$ ,*

$$\text{ad}_u^* m = -P[(u \cdot \nabla)m + (\nabla u)^\top m], \quad ((\nabla u)^\top m)_i = m_j \partial_i u^j.$$

*Proof.* By definition  $\langle \text{ad}_u^* m, v \rangle = \langle m, [u, v] \rangle$  for  $v \in \mathfrak{g}$ , and

$$\langle m, [u, v] \rangle = \int m_i (u^j \partial_j v^i - v^j \partial_j u^i) = - \int (u^j \partial_j m_i) v^i - \int m_j (\partial_i u^j) v^i,$$

integrating the first term by parts and using  $\text{div } u = 0$ , and relabelling indices in the second. As this holds for all divergence-free  $v$ , the formula follows after applying  $P$ .  $\square$

So (2) is the system

$$\partial_t m + (u \cdot \nabla)m + (\nabla u)^\top m = -\nabla p, \quad u = A^{-1}m, \quad \text{div } u = 0. \quad (3)$$

**Proposition 2.2** (the reduction). *Let  $u$  be divergence-free and let  $m$  be any vector field. With  $w = \text{curl } m$ ,*

$$\text{curl}[(u \cdot \nabla)m + (\nabla u)^\top m] = (u \cdot \nabla)w - (w \cdot \nabla)u.$$

*Consequently a solution of (3) satisfies*

$$\partial_t w + (u \cdot \nabla)w = (w \cdot \nabla)u, \quad \text{div } w = 0, \quad (4)$$

*for every  $s$ : the vortex-stretching equation of the unmodified Euler equation.*

*Proof.* Regard  $m$  as a 1-form; in Cartesian coordinates  $(\mathcal{L}_u m)_i = u^j \partial_j m_i + m_j \partial_i u^j$ , the bracketed expression. Since  $d$  commutes with  $\mathcal{L}_u$  we have  $\mathcal{L}_u(dm) = d(\mathcal{L}_u m)$ . Identify the 2-form  $dm$  with the vector field  $w = \text{curl } m$  by  $dm = \iota_w \text{vol}$ . Then

$$\mathcal{L}_u(\iota_w \text{vol}) = \iota_{[u, w]} \text{vol} + \iota_w \mathcal{L}_u \text{vol} = \iota_{[u, w]} \text{vol},$$

since  $\mathcal{L}_u \text{vol} = (\text{div } u) \text{vol} = 0$ , and  $[u, w] = (u \cdot \nabla)w - (w \cdot \nabla)u$ . Taking curl of (3) kills the gradient and gives (4).  $\square$

*Remark 2.3.* The force of Proposition 2.2 is its quantifier: the identity holds for arbitrary  $m$ , so the transported quantity obeys Euler’s own equation whichever inertia operator is used, and nothing is lost or gained there. What varies with  $s$  is only the constitutive law recovering  $u$  from  $w$ . This is why the family is tractable by Euler’s own methods, and why the criterion below is a statement about a kernel rather than about a dynamics.

**Lemma 2.4** (the orders). *On mean-zero divergence-free fields,  $\hat{m}(k) = i|k|^{-2} k \times \hat{w}(k)$  and*

$$\hat{u}(k) = \frac{i}{|k|^{2+2s}} k \times \hat{w}(k), \quad \widehat{\partial_j u^i}(k) = -\frac{k_j (k \times \hat{w}(k))_i}{|k|^{2+2s}}.$$

*Thus  $u = S_s w$  and  $\nabla u = T_s w$  for Fourier multipliers whose symbols are smooth away from the origin and homogeneous of degrees  $-1 - 2s$  and  $-2s$  respectively. In particular, for every  $\sigma$ ,*

$$\|u\|_{H^\sigma} \lesssim \|w\|_{H^{\sigma-1-2s}}, \quad \|\nabla u\|_{H^\sigma} \lesssim \|w\|_{H^{\sigma-2s}}. \quad (5)$$

*Proof.* Since  $\hat{m}(k) \perp k$ ,  $k \times \hat{w} = ik \times (k \times \hat{m}) = i(k(k \cdot \hat{m}) - |k|^2 \hat{m}) = -i|k|^2 \hat{m}$ , which inverts to the stated formula; then  $\hat{u} = a(|k|)^{-1} \hat{m} = |k|^{-2s} \hat{m}$  and  $\widehat{\partial_j u^i} = ik_j \hat{u}^i$ . The symbol of  $T_s$  is a homogeneous polynomial of degree 2 over  $|k|^{2+2s}$ .  $\square$

**Lemma 2.5** (the conserved quantity). *For  $u \in \mathfrak{g}$  and  $w = \text{curl } Au$ ,*

$$E = \langle Au, u \rangle = \|u\|_{\dot{H}^s}^2 = \|w\|_{\dot{H}^{-1-s}}^2,$$

*and  $E$  is conserved along (2).*

*Proof.*  $|\hat{w}(k)| = |k| |\hat{m}(k)|$  because  $\hat{m}(k) \perp k$ , and  $\hat{m} = |k|^{2s} \hat{u}$ , so  $|k|^{-2-2s} |\hat{w}|^2 = |k|^{-2s} |\hat{m}|^2 = |k|^{2s} |\hat{u}|^2$ ; sum over  $k \neq 0$ . Conservation is the Lie–Poisson cancellation:  $\frac{d}{dt} \langle m, A^{-1} m \rangle = 2 \langle A^{-1} m, \dot{m} \rangle = 2 \langle u, \text{ad}_u^* m \rangle = 2 \langle m, [u, u] \rangle = 0$ .  $\square$

### 3. THE KERNEL ESTIMATE

**Lemma 3.1.** *Let  $s > 0$  and let  $T_s$  be as in Lemma 2.4. Let  $p \in [1, \infty]$  satisfy  $p > 3/(2s)$ . Then for mean-zero  $w \in L^p(\mathbb{T}^3)$ ,*

$$\|T_s w\|_{L^\infty} \leq C(s, p) \|w\|_{L^p}.$$

*Proof.* The symbol of  $T_s$  is smooth on  $\mathbb{R}^3 \setminus \{0\}$  and homogeneous of degree  $-2s$ . The convolution kernel on  $\mathbb{T}^3$  of such a multiplier agrees, modulo a function smooth on all of  $\mathbb{T}^3$ , with the periodisation of the kernel of the corresponding multiplier on  $\mathbb{R}^3$ , which for degree  $-2s$  with  $0 < 2s < 3$  is homogeneous of degree  $2s - 3$  [16, Ch. III, V]. Hence the periodic kernel  $\kappa_s$  obeys  $|\kappa_s(x)| \leq C|x|^{2s-3}$  for  $0 < |x| \leq \pi$  and is bounded elsewhere. For  $2s > 3$  the kernel is bounded and continuous, and for  $2s = 3$  it is logarithmic.

For  $0 < 2s < 3$  we have  $\kappa_s \in L^{p'}(\mathbb{T}^3)$  if and only if  $(3 - 2s)p' < 3$ , that is  $p' < 3/(3 - 2s)$ , that is  $p > 3/(2s)$ ; Young's inequality  $\|\kappa_s * w\|_{L^\infty} \leq \|\kappa_s\|_{L^{p'}} \|w\|_{L^p}$  gives the claim. For  $2s \geq 3$  the kernel lies in every  $L^q$ ,  $q < \infty$ , so every  $p > 1$  is admissible, and for  $2s > 3$  boundedness admits  $p = 1$ .  $\square$

*Remark 3.2* (numerical check). The two quantitative claims in the proof are checked by `kernel_check.py`, for the model symbol  $|k|^{-2s}$  and for  $k_1 k_2 / |k|^{2+2s}$ , which is the shape of an entry of  $T_s$ . The integrability threshold is confirmed sharply: for  $s \in \{0.25, 0.5, 1, 1.25\}$  the discrete  $L^{p'}$  norm of the kernel converges under grid refinement at  $p' = 0.8p'_c$  and grows at  $p' = 1.2p'_c$ , with  $p'_c = 3/(3 - 2s)$  in each case. The pointwise decay is checked by summing the Fourier series along a ray with a smooth cutoff at  $|k| = K$ ; the fitted exponent approaches  $2s - 3$  monotonically in  $K$ , the gap at  $K = 64$  being 0.34, 0.15, 0.066, 0.027 and 0.006 for  $s = 0.25, \dots, 1.25$ . The slower convergence at small  $s$  is expected, the kernel being more singular there and needing more modes to resolve.

*Remark 3.3* (why  $s = 0$  differs in kind). At  $s = 0$  the symbol of  $T_0$  is homogeneous of degree 0 and  $T_0$  is a Calderón–Zygmund operator, unbounded from  $L^\infty$  to  $L^\infty$ . Lemma 3.1 then fails for every  $p$ , and the classical route substitutes the logarithmic Sobolev inequality of Brezis–Gallouët–Wainger [4, 5], in the form  $\|\nabla u\|_{L^\infty} \lesssim \|\omega\|_{L^\infty} (1 + \log(e + \|u\|_{H^r})) + \|\omega\|_{L^2}$ , which is precisely the logarithm in [3].

### 4. THE A PRIORI ESTIMATE

Throughout this section  $s > 0$  and  $w$  is a smooth solution of (4) on  $[0, T)$  with  $u = S_s w$  as in Lemma 2.4. We use two standard estimates. The first is the Kato–Ponce commutator estimate [13]: for  $\sigma > 0$ ,

$$\|[\Lambda^\sigma, f]g\|_{L^2} \lesssim \|\nabla f\|_{L^\infty} \|g\|_{H^{\sigma-1}} + \|f\|_{H^\sigma} \|g\|_{L^\infty}. \quad (6)$$

The second is the Moser product estimate [14, Ch. 3]:  $\|fg\|_{H^\sigma} \lesssim \|f\|_{L^\infty} \|g\|_{H^\sigma} + \|f\|_{H^\sigma} \|g\|_{L^\infty}$ .

**Lemma 4.1** (commutator with a divergence-free drift). *Let  $r > 0$  and let  $u$  be divergence-free. Then*

$$\|[\Lambda^r, u \cdot \nabla]w\|_{L^2} \lesssim \|\nabla u\|_{L^\infty} \|w\|_{H^r} + \|u\|_{H^{r+1}} \|w\|_{L^\infty}.$$

*Proof.* Write  $[\Lambda^r, u]w$  for the array with entries  $[\Lambda^r, u^j]w$ . Since  $\text{div } u = 0$  we have  $\partial_j(u^j w) = u^j \partial_j w$ , and therefore

$$\text{div}([\Lambda^r, u]w) = \sum_j \partial_j (\Lambda^r(u^j w) - u^j \Lambda^r w) = \Lambda^r(u \cdot \nabla w) - u \cdot \nabla \Lambda^r w = [\Lambda^r, u \cdot \nabla]w.$$

The commutator on the left carries no derivative of  $w$ ; that is the whole point. Since  $\|\text{div } F\|_{L^2} \leq \|\Lambda F\|_{L^2}$ ,

$$\|[\Lambda^r, u \cdot \nabla]w\|_{L^2} \leq \|\Lambda[\Lambda^r, u]w\|_{L^2} \leq \|[\Lambda^{r+1}, u]w\|_{L^2} + \|[\Lambda, u]\Lambda^r w\|_{L^2},$$

using the identity  $\Lambda[\Lambda^r, u]w = [\Lambda^{r+1}, u]w - [\Lambda, u]\Lambda^r w$ , which is immediate on expanding both sides. For the first term, (6) at order  $r+1$  gives  $\|\nabla u\|_{L^\infty} \|w\|_{H^r} + \|u\|_{H^{r+1}} \|w\|_{L^\infty}$ . For the second, Calderón's commutator theorem [9] gives  $\|[\Lambda, u]h\|_{L^2} \lesssim \|\nabla u\|_{L^\infty} \|h\|_{L^2}$ , applied with  $h = \Lambda^r w$ .  $\square$

**Lemma 4.2** (transport bound). *For every  $q \in [1, \infty]$  and  $t < T$ ,*

$$\|w(t)\|_{L^q} \leq \|w_0\|_{L^q} \exp\left(\int_0^t \|\nabla u(\tau)\|_{L^\infty} d\tau\right).$$

*Proof.* Along the flow of  $u$ , (4) gives  $\frac{d}{dt}|w| \leq |\nabla u| |w|$  pointwise, and the flow preserves Lebesgue measure because  $\text{div } u = 0$ ; integrate.  $\square$

**Proposition 4.3** (a priori estimate). *Let  $s > 0$  and  $r > \frac{5}{2}$ . Then, with  $C = C(r, s)$ ,*

$$\frac{d}{dt} \|w\|_{H^r} \leq C(\|\nabla u\|_{L^\infty} + \|w\|_{L^\infty}) \|w\|_{H^r}.$$

*Proof.* Pair  $\Lambda^r$  of (4) with  $\Lambda^r w$ :

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^r}^2 = -\langle [\Lambda^r, u \cdot \nabla]w, \Lambda^r w \rangle - \langle u \cdot \nabla \Lambda^r w, \Lambda^r w \rangle + \langle \Lambda^r(w \cdot \nabla u), \Lambda^r w \rangle.$$

The middle term vanishes because  $\text{div } u = 0$ .

*Stretching term.* By the Moser estimate and then (5),

$$\|\Lambda^r(w \cdot \nabla u)\|_{L^2} \lesssim \|w\|_{L^\infty} \|\nabla u\|_{H^r} + \|w\|_{H^r} \|\nabla u\|_{L^\infty} \lesssim (\|w\|_{L^\infty} + \|\nabla u\|_{L^\infty}) \|w\|_{H^r},$$

using  $\|\nabla u\|_{H^r} \lesssim \|w\|_{H^{r-2s}} \leq \|w\|_{H^r}$  for  $s > 0$ .

*Commutator term.* By Lemma 4.1 and then (5),

$$\|[\Lambda^r, u \cdot \nabla]w\|_{L^2} \lesssim \|\nabla u\|_{L^\infty} \|w\|_{H^r} + \|u\|_{H^{r+1}} \|w\|_{L^\infty} \lesssim (\|\nabla u\|_{L^\infty} + \|w\|_{L^\infty}) \|w\|_{H^r},$$

since  $\|u\|_{H^{r+1}} \lesssim \|w\|_{H^{r-2s}} \leq \|w\|_{H^r}$ , again by (5) and  $s > 0$ .

Both contributions are thus at most  $C(\|\nabla u\|_{L^\infty} + \|w\|_{L^\infty}) \|w\|_{H^r}^2$ ; dividing by  $\|w\|_{H^r}$  gives the statement.  $\square$

*Remark 4.4* (no threshold in  $r$ ). The estimate holds for every  $r > \frac{5}{2}$ , with no threshold depending on  $s$ , and its proof uses no interpolation. What buys this is the divergence form in Lemma 4.1: writing the commutator as  $\text{div}([\Lambda^r, u]w)$  moves the derivative off  $w$  and onto  $u$ , and by (5)  $u$  has  $1 + 2s$  derivatives to spare, of which the manoeuvre spends one. Applying (6) directly instead leaves  $\|u\|_{H^r} \|\nabla w\|_{L^\infty}$ , whose second factor has then to be interpolated against  $\|w\|_{H^r}$ ; that route closes only for  $r$  above an  $s$ -dependent threshold which diverges as  $s \rightarrow 0^+$ , and is an artefact of the manoeuvre rather than of the equation. At  $s = 0$  nothing is left over, the two routes coincide, and the estimate above is the classical one for the Euler equation.

Uniqueness is not assumed; it follows from the smoothing of  $S_s$  and  $T_s$  by an  $L^2$  argument.

**Proposition 4.5** (uniqueness). *Let  $s > 0$ ,  $r > \frac{5}{2}$  and  $T > 0$ . If  $w_1, w_2 \in C([0, T]; H^r)$  both solve (4) with  $w_1(0) = w_2(0)$ , then  $w_1 \equiv w_2$ .*

*Proof.* Put  $\delta = w_1 - w_2$  and  $v = u_1 - u_2 = S_s \delta$ . Subtracting the two copies of (4),

$$\partial_t \delta + (u_1 \cdot \nabla) \delta + (v \cdot \nabla) w_2 = (\delta \cdot \nabla) u_1 + (w_2 \cdot \nabla) v.$$

Pair with  $\delta$  in  $L^2$ . The term  $\langle (u_1 \cdot \nabla) \delta, \delta \rangle$  vanishes since  $\operatorname{div} u_1 = 0$ . Because  $s > 0$ , the symbols  $|k|^{-1-2s}$  and  $|k|^{-2s}$  of  $S_s$  and  $T_s$  are bounded by 1 on  $|k| \geq 1$ , so  $\|v\|_{L^2} \leq \|\delta\|_{L^2}$  and  $\|\nabla v\|_{L^2} = \|T_s \delta\|_{L^2} \leq \|\delta\|_{L^2}$ . Hence

$$|\langle (v \cdot \nabla) w_2, \delta \rangle| \leq \|\nabla w_2\|_{L^\infty} \|\delta\|_{L^2}^2, \quad |\langle (\delta \cdot \nabla) u_1, \delta \rangle| \leq \|\nabla u_1\|_{L^\infty} \|\delta\|_{L^2}^2,$$

$$|\langle (w_2 \cdot \nabla) v, \delta \rangle| \leq \|w_2\|_{L^\infty} \|\nabla v\|_{L^2} \|\delta\|_{L^2} \leq \|w_2\|_{L^\infty} \|\delta\|_{L^2}^2.$$

Since  $r > \frac{5}{2}$ , the three coefficients are bounded on  $[0, T]$  by  $C(\|w_1\|_{C([0, T]; H^r)} + \|w_2\|_{C([0, T]; H^r)})$ . Grönwall with  $\delta(0) = 0$  gives  $\delta \equiv 0$ .  $\square$

## 5. LOCAL EXISTENCE

It remains to produce a solution. We do so by Galerkin truncation, which on  $\mathbb{T}^3$  is cleaner than mollification: the Fourier projection is self-adjoint, idempotent, and commutes with every operator appearing in the problem. Solutions are taken in the class

**Definition 5.1.** For  $T > 0$  and  $r > \frac{5}{2}$  set

$$\mathcal{X}_T^r := L^\infty([0, T]; H^r) \cap \bigcap_{\frac{5}{2} < r' < r} C([0, T]; H^{r'}),$$

and call  $w \in \mathcal{X}_T^r$  a *solution* of (4) if the equation holds in  $C([0, T]; H^{r'-1})$  for every such  $r'$ .

Every statement above holds in this class: Propositions 4.3 and 4.5 and Lemma 4.2 use only the size of  $\|w(t)\|_{H^r}$ , not continuity in  $H^r$ . Membership of  $H^r$  at *every* time, rather than almost every time, is not free, and is what the continuation argument of Section 6 needs; it follows from the class.

**Lemma 5.2** (weak continuity). *Let  $w \in \mathcal{X}_T^r$  and let  $M$  be the essential supremum of  $\|w(t)\|_{H^r}$  on  $[0, T]$ . Then  $w(t) \in H^r$  for every  $t \in [0, T]$ , with  $\sup_{[0, T]} \|w(t)\|_{H^r} = M$ , and  $w : [0, T] \rightarrow H^r$  is weakly continuous.*

*Proof.* Fix  $t$  and choose  $t_n \rightarrow t$  with  $\|w(t_n)\|_{H^r} \leq M$ ; this is possible since the set of such times has full measure. Fix  $r' \in (\frac{5}{2}, r)$ . The sequence  $(w(t_n))$  is bounded in  $H^r$ , so a subsequence converges weakly in  $H^r$  to some  $z$ , and weak convergence in  $H^r$  implies weak convergence in  $H^{r'}$ . As  $w \in C([0, T]; H^{r'})$  we also have  $w(t_n) \rightarrow w(t)$  strongly, hence weakly, in  $H^{r'}$ . Therefore  $z = w(t)$ , so  $w(t) \in H^r$ , and weak lower semicontinuity of the norm gives  $\|w(t)\|_{H^r} \leq M$ . The reverse inequality is immediate, so the supremum equals  $M$ .

For weak continuity, let  $\psi$  be a trigonometric polynomial. Then  $\langle \Lambda^r w(t), \Lambda^r \psi \rangle = \langle w(t), \Lambda^{2r} \psi \rangle$  is continuous in  $t$  because  $w \in C([0, T]; L^2)$ . Such  $\psi$  are dense in  $H^r$  and  $\|w(t)\|_{H^r} \leq M$  uniformly, so  $t \mapsto w(t)$  is weakly continuous into  $H^r$ .  $\square$

A solution in  $\mathcal{X}_T^r$  is in fact *strongly* continuous into  $H^r$ . That is the argument of Bona and Smith [7], which applies here unchanged — the equation (4) is reversible,  $-w(-t)$  solving it whenever  $w(t)$  does, which is what the argument uses to pass from continuity of the norm at  $t = 0$  to continuity at every time. We omit it because no statement in this paper invokes it: Lemma 5.2 supplies everything used below.

**Theorem 5.3** (local existence). *Let  $s > 0$  and  $r > \frac{5}{2}$ . There is  $C = C(r, s)$  such that every mean-zero divergence-free  $w_0 \in H^r(\mathbb{T}^3)$ ,  $w_0 \neq 0$ , admits a solution  $w \in \mathcal{X}_T^r$  of (4) with  $w(0) = w_0$  on  $[0, T]$ , where*

$$T = \tau(\|w_0\|_{H^r}), \quad \tau(\rho) := \frac{1}{2C\rho},$$

and moreover  $\sup_{[0, T]} \|w\|_{H^r} \leq 2\|w_0\|_{H^r}$ . The function  $\tau$  is non-increasing.

*Proof.* Write  $B(w) := -[u, w] = -(u \cdot \nabla)w + (w \cdot \nabla)u$  with  $u = S_s w$ , so that (4) reads  $\partial_t w = B(w)$ , and put  $\rho := \|w_0\|_{H^r}$ .

*Step 1: the truncated system.* For  $N \in \mathbb{N}$  let  $P_N$  be the Fourier projection onto  $\{0 < |k| \leq N\}$  and  $V_N := P_N \mathfrak{g}$ , a finite-dimensional space of mean-zero divergence-free trigonometric fields. Consider

$$\partial_t w^N = P_N B(w^N), \quad w^N(0) = P_N w_0. \quad (7)$$

The right-hand side maps  $V_N$  to  $V_N$ :  $B(w)$  is a Lie bracket of divergence-free fields, hence divergence-free, and has zero mean because  $\int [u, w] = \int ((\text{div } w)u - (\text{div } u)w) = 0$ . It is a quadratic polynomial map of the finite-dimensional space  $V_N$ , so Picard–Lindelöf gives a unique maximal solution on some  $[0, T_N)$ .

*Step 2: a bound uniform in  $N$ .* Since  $P_N$  is self-adjoint, commutes with  $\Lambda^r$ , and fixes  $w^N$ ,

$$\frac{1}{2} \frac{d}{dt} \|w^N\|_{H^r}^2 = \langle \Lambda^r P_N B(w^N), \Lambda^r w^N \rangle = \langle \Lambda^r B(w^N), \Lambda^r P_N w^N \rangle = \langle \Lambda^r B(w^N), \Lambda^r w^N \rangle,$$

so the truncation is invisible and the proof of Proposition 4.3 applies to  $w^N$  verbatim, with the same constant. Each term of the bracket there is  $\lesssim \|w^N\|_{H^r}$ : indeed  $\|w\|_{L^\infty} \lesssim \|w\|_{H^r}$  since  $r > \frac{3}{2}$ ;  $\|\nabla u\|_{L^\infty} = \|T_s w\|_{L^\infty} \lesssim \|T_s w\|_{H^2} \lesssim \|w\|_{H^{2-2s}} \leq \|w\|_{H^r}$  by (5), using  $s > 0$  and  $r > 2$ . Hence

$$\frac{d}{dt} \|w^N\|_{H^r} \leq C \|w^N\|_{H^r}^2. \quad (8)$$

As  $\|w^N(0)\|_{H^r} \leq \rho$ , integrating (8) gives  $\|w^N(t)\|_{H^r} \leq \rho/(1 - C\rho t) \leq 2\rho$  for  $t \leq T = 1/(2C\rho)$ , uniformly in  $N$ . All norms on  $V_N$  are equivalent, so this bound also forbids the ODE from leaving every ball, whence  $T_N > T$  and  $w^N$  exists on  $[0, T]$ .

*Step 3: Cauchy in  $L^2$ .* Let  $M > N$  and  $\delta := w^M - w^N$ , so  $P_M \delta = \delta$  and

$$\partial_t \delta = P_M [B(w^M) - B(w^N)] - (P_M - P_N)B(w^N).$$

Pair with  $\delta$  and use  $\langle P_M F, \delta \rangle = \langle F, \delta \rangle$ . The first term is bounded exactly as in the proof of Proposition 4.5 — that argument used only that both fields lie in  $H^r$  with  $r > \frac{5}{2}$  — by  $C(\|w^M\|_{H^r} + \|w^N\|_{H^r}) \|\delta\|_{L^2}^2 \leq C\rho \|\delta\|_{L^2}^2$ . For the second,  $H^{r-1}$  is an algebra because  $r-1 > \frac{3}{2}$ , so by (5)  $\|B(w^N)\|_{H^{r-1}} \lesssim \|u^N\|_{H^r} \|w^N\|_{H^r} \lesssim \rho^2$ , while  $P_M - P_N$  is supported in  $|k| > N$ , whence  $\|(P_M - P_N)B(w^N)\|_{L^2} \leq N^{-(r-1)} \|B(w^N)\|_{H^{r-1}} \lesssim \rho^2 N^{-(r-1)}$ . Therefore

$$\frac{d}{dt} \|\delta\|_{L^2} \leq C\rho \|\delta\|_{L^2} + C\rho^2 N^{-(r-1)},$$

and  $\|\delta(0)\|_{L^2} = \|(P_M - P_N)w_0\|_{L^2} \leq N^{-r}\rho$ , so Grönwall gives  $\sup_{[0, T]} \|\delta\|_{L^2} \leq C(\rho, T) N^{-(r-1)}$  uniformly in  $M$ . Thus  $(w^N)$  is Cauchy in  $C([0, T]; L^2)$ .

*Step 4: passage to the limit.* Let  $w$  be the limit in  $C([0, T]; L^2)$ . For  $r' < r$ , interpolation and Step 2 give  $\|w^N - w^M\|_{H^{r'}} \leq \|w^N - w^M\|_{L^2}^{1-r'/r} \|w^N - w^M\|_{H^r}^{r'/r}$ , so  $(w^N)$  is Cauchy in  $C([0, T]; H^{r'})$  and  $w \in C([0, T]; H^{r'})$ . Weak-\* compactness with lower semicontinuity of the norm gives  $w \in L^\infty([0, T]; H^r)$  with  $\sup_{[0, T]} \|w\|_{H^r} \leq 2\rho$ . Fix  $r' \in (\frac{5}{2}, r)$ . Then  $H^{r'-1}$  is an algebra and, by (5),  $B$  is locally Lipschitz from  $H^{r'}$  to  $H^{r'-1}$ , so  $B(w^N) \rightarrow B(w)$  in  $C([0, T]; H^{r'-1})$ . Also  $P_N \rightarrow I$  strongly on  $H^{r'-1}$ , hence uniformly on the compact set  $\{B(w(t)) : t \in [0, T]\}$ .

Passing to the limit in (7) gives  $\partial_t w = B(w)$  in  $C([0, T]; H^{r'-1})$ . Hence  $w \in \mathcal{X}_T^r$  solves (4) with  $w(0) = w_0$ .  $\square$

*Remark 5.4.* The scheme is the classical one [14, Ch. 3]. What lets it run without modification is that  $S_s$  and  $T_s$  are *smoother* than their Euler counterparts by  $2s$  derivatives, so every estimate above has room to spare; nothing in the argument is sharp in  $s$ , and none of it would survive  $s < 0$ , where (5) reverses.

## 6. THE CRITERION

**Proposition 6.1** (maximal solution and the blowup alternative). *Let  $s > 0$  and  $r > \frac{5}{2}$ . Each mean-zero divergence-free  $w_0 \in H^r$  admits a unique maximal solution on some  $[0, T_*)$ ,  $T_* \in (0, \infty]$ , and if  $T_* < \infty$  then*

$$\sup_{t < T_*} \|w(t)\|_{H^r} = \infty.$$

*Proof.* By Proposition 4.5 any two solutions with the same datum agree on the intersection of their intervals, so the union of all solutions is a well-defined solution on a maximal  $[0, T_*)$ . Suppose  $T_* < \infty$  and  $M := \sup_{t < T_*} \|w(t)\|_{H^r} < \infty$ . Pick  $t_0 < T_*$  with  $T_* - t_0 < \tau(M)$ , which is possible as  $\tau(M) > 0$ . By Lemma 5.2 the datum  $w(t_0)$  lies in  $H^r$  — at that particular time, not merely almost everywhere — with  $\|w(t_0)\|_{H^r} \leq M$ , so Theorem 5.3 and the monotonicity of  $\tau$  give a solution on  $[t_0, t_0 + \tau(M)]$ , and by Proposition 4.5 it extends  $w$  past  $T_*$ . This contradicts maximality.  $\square$

**Theorem 6.2** (continuation in  $\|\nabla u\|_{L^\infty}$ ). *Let  $s > 0$  and  $r > \frac{5}{2}$ . If  $T < \infty$  and*

$$\int_0^T \|\nabla u(t)\|_{L^\infty} dt < \infty,$$

*then  $\sup_{t < \min(T, T_*)} \|w(t)\|_{H^r} < \infty$ ; in particular  $T < T_*$ .*

*Proof.* Write  $I := \int_0^T \|\nabla u\|_{L^\infty} dt < \infty$ . By Lemma 4.2,  $\|w(t)\|_{L^2}$  and  $\|w(t)\|_{L^\infty}$  are bounded on  $[0, \min(T, T_*)]$  by  $\|w_0\|_{L^2} e^I$  and  $\|w_0\|_{L^\infty} e^I$ . Hence the bracket in Proposition 4.3 is bounded by  $\|\nabla u(t)\|_{L^\infty} + M$  for a constant  $M$  depending only on  $\|w_0\|_{L^2}$ ,  $\|w_0\|_{L^\infty}$ ,  $I$ ,  $r$  and  $s$ . Grönwall gives

$$\|w(t)\|_{H^r} \leq \|w_0\|_{H^r} e^{C(I+MT)},$$

which is finite. By the blowup alternative of Proposition 6.1 this is incompatible with  $T_* \leq T$ .  $\square$

**Theorem 6.3** (the criterion). *Let  $s > 0$ ,  $r > \frac{5}{2}$ , and let  $p \in [1, \infty]$  satisfy  $p > 3/(2s)$ . If  $T < \infty$  and*

$$\int_0^T \|w(t)\|_{L^p} dt < \infty,$$

*then the solution does not break down at or before  $T$ .*

*Proof.* By Lemma 3.1,  $\|\nabla u(t)\|_{L^\infty} \leq C(s, p) \|w(t)\|_{L^p}$  for each  $t$ , so

$$\int_0^T \|\nabla u\|_{L^\infty} dt \leq C(s, p) \int_0^T \|w\|_{L^p} dt < \infty,$$

and Theorem 6.2 applies.  $\square$

*Remark 6.4.* Only  $p$  is constrained by  $s$ . As  $s \rightarrow 0^+$  the admissible exponent  $p > 3/(2s)$  escapes to  $\infty$  and no finite  $p$  survives the limit, which is the correct behaviour: at  $s = 0$  the estimate of Lemma 3.1 is false for every  $p$  (Remark 3.3) and the logarithm is unavoidable. The regularity requirement is  $r > \frac{5}{2}$  throughout, independently of  $s$ , by Remark 4.4.

## 7. WHAT THE CRITERION DOES NOT DO

**It recovers  $5/2$  and moves nothing.** By Lemma 2.5 the conserved quantity is  $\|w\|_{\dot{H}^{-1-s}}$ . By Lemma 2.4 and Sobolev embedding,  $\|\nabla u\|_{L^\infty} = \|T_s w\|_{L^\infty} \lesssim \|w\|_{\dot{H}^{3/2-2s+\varepsilon}}$  for every  $\varepsilon > 0$ , and the conserved norm dominates this exactly when

$$\frac{3}{2} - 2s \leq -1 - s \iff s \geq \frac{5}{2},$$

the geodesic completeness threshold of [8] and, as  $s > \frac{1}{2} \dim M + 1$ , the same count that returns  $3/2$  on  $\text{Diff}(S^1)$  [12, 2]. So Theorem 6.3 is consistent with the known completeness result and improves on it nowhere.

That is the outcome to insist on. A continuation criterion improving the threshold by itself would, continued to  $s = 0$ , prove global regularity for the three-dimensional Euler equation. What the criterion does is make the problem definite: bound  $\|w\|_{L^p}$  for a single  $p > 3/(2s)$ , given only that  $\|w\|_{\dot{H}^{-1-s}}$  is conserved and that  $w$  obeys (4). Since  $-1 - s < 0$  and negative-regularity spaces do not embed in  $L^p$  for  $p \geq 2$ , the conservation law cannot supply the hypothesis directly at any  $s < 5/2$ , and the gain must come from the equation.

**The threshold is the critical exponent, not an artefact.** It is worth saying why  $5/2$  is where it is, because the answer removes a hope. Equation (4) admits the dilations  $w_\lambda(x, t) = \lambda^a w(\lambda x, \lambda^b t)$ . As  $S_s$  is homogeneous of degree  $-1 - 2s$  we get  $u_\lambda = \lambda^{a-1-2s} u(\lambda x, \lambda^b t)$ , so both nonlinear terms carry  $\lambda^{2a-2s}$  while  $\partial_t w$  carries  $\lambda^{a+b}$ , and the equation is invariant precisely when  $b = a - 2s$ . In dimension  $d$  one has  $\|\lambda^a f(\lambda \cdot)\|_{\dot{H}^\sigma} = \lambda^{a-d/2+\sigma} \|f\|_{\dot{H}^\sigma}$ , so by Lemma 2.5 the conserved energy  $\|w\|_{\dot{H}^{-1-s}}^2$  is scale-invariant exactly at  $a = \frac{d}{2} + 1 + s$ , and then

$$b = \frac{d}{2} + 1 - s.$$

The energy-invariant dilation therefore concentrates in finite time when  $s < \frac{d}{2} + 1$  and spreads when  $s > \frac{d}{2} + 1$ . That is,

$$s^* = \frac{d}{2} + 1$$

is the scaling-critical exponent. In  $d = 3$  this is  $5/2$ , the threshold of [8]; in  $d = 1$  it is  $3/2$ , which is the *sharp* threshold on  $\text{Diff}(S^1)$  [12, 2]. So on the circle, where the answer is known, completeness holds exactly in the subcritical range and breakdown is known, and conjectured throughout, exactly in the supercritical range.

Two consequences, neither of them the obvious one. First, the coincidence of the count of the previous subsection with  $\frac{d}{2} + 1$  is no accident of the method — but it is a statement about the *energy*, not about the problem. The energy is scale-invariant exactly at  $s^*$ , so below  $s^*$  it is blind to concentration and no argument resting on it alone can reach there. The same count puts the criterion of Theorem 6.3 on the other side of the line:  $\|\lambda^a w(\lambda \cdot)\|_{L^p} = \lambda^{a-d/p} \|w\|_{L^p}$  against  $\|\nabla u\|_{L^\infty} \sim \lambda^{a-2s}$ , so  $L^p$  is subcritical exactly when  $p > d/(2s)$  — which is precisely the range Lemma 3.1 admits,  $3/(2s)$  in three dimensions and  $1/s$  in two. The criterion is therefore stated in a subcritical quantity, and the question is whether any such quantity can be controlled. Both counts are checked in `scaling_check.py`.

In two dimensions one can be, and the whole supercritical range falls.

**Proposition 7.1** (the two-dimensional case). *Let  $d = 2$  and  $s > 0$ . The geodesic flow of the metric (1) on  $\text{SDiff}(\mathbb{T}^2)$  is globally regular: for every  $r > \frac{5}{2}$  and every mean-zero divergence-free  $w_0 \in H^r(\mathbb{T}^2)$  the solution exists for all time.*

*Proof.* Sections 2–6 go through in  $d = 2$  with two changes. First, in the proof of Proposition 2.2 the field  $w = \text{curl } m$  is a scalar and  $dm$  a 2-form, so  $\mathcal{L}_u(w \text{ vol}) = (u \cdot \nabla w) \text{ vol}$  for divergence-free

$u$ , and the reduction reads

$$\partial_t w + (u \cdot \nabla)w = 0$$

with *no* stretching term — there being no way to form  $(w \cdot \nabla)u$  out of a scalar. The transport is by a measure-preserving flow, so  $\|w(t)\|_{L^p} = \|w_0\|_{L^p}$  for every  $p$  and every  $t$ . Second, the kernel in Lemma 3.1 is homogeneous of degree  $2s - 2$  and lies in  $L^{p'}(\mathbb{T}^2)$  iff  $(2 - 2s)p' < 2$ , that is iff  $p > 1/s$ . Fixing such a  $p$ ,

$$\int_0^T \|\nabla u\|_{L^\infty} dt \leq C(s, p) T \|w_0\|_{L^p} < \infty$$

for every  $T$ , and Theorem 6.2, whose proof is unchanged, forbids breakdown at  $T$ . As  $T$  was arbitrary the solution is global.  $\square$

*Remark 7.2.* The argument is elementary once the reduction is in hand and we claim no novelty for it; its role here is comparative, and it is worth three observations. It fails at  $s = 0$ , where Lemma 3.1 is unavailable for every  $p$  and one is returned to the classical two-dimensional theory and its logarithm. Since  $s^* = 2$  in two dimensions, it covers the whole supercritical range  $[0, 2)$ : on  $\text{SDiff}(\mathbb{T}^2)$  volume preservation buys every order below criticality, so scale-criticality bounds what the energy can do and not what the equation does. And the mechanism is exactly the one absent in three dimensions — the subcritical  $L^p$  norms are conserved there because nothing stretches. The identity underlying the reduction is checked in `dim2_check.py`.

It is tempting, separately, to read the one-dimensional case as evidence that the threshold sits *at* criticality — on  $\text{Diff}(S^1)$  it does, exactly — and to expect the same in three dimensions. That reading is wrong.  $\text{Diff}(S^1)$  is the full diffeomorphism group, and the comparison bearing on  $\text{SDiff}(\mathbb{T}^3)$  is between  $\text{Diff}$  and  $\text{SDiff}$  in a *fixed* dimension; carried out in  $d = 2$  by Proposition 7.1, against the threshold  $s^*$  which  $\text{Diff}$  attains in  $d = 1$  [12, 2] and in  $d \geq 3$  [6, 8], it goes the other way.

The two analogies thus point in opposite directions, and three dimensions is undetermined by either. What distinguishes it is that the stretching term neither vanishes, as in  $d = 2$ , nor acts unimpeded, as on  $\text{Diff}$ : it is present but depleted, by the alignment cancellation of Lemma 7.3, which is the three-dimensional residue of the two-dimensional vanishing. How much that depletion is worth admits a definite answer, in two parts, and neither is a reserve of unused strength.

*Unconditionally it is worth nothing.* Theorem 6.3 already rests on the bound  $|\kappa_s(y)| \leq C|y|^{2s-3}$  of Lemma 3.1, whose exponent is sharp (Remark 3.2). The alignment factor multiplies that kernel by  $|D_s| \leq \min\{C, C|\xi' - \xi|\}$ , and absent a hypothesis on  $\xi$  the second bound is no improvement on the first,  $|\xi' - \xi| \leq 2$  always: a direction field may turn through a right angle on the scale at which  $|w|$  varies, and then nothing is depleted at all. Measured, the effect is starker than that argument suggests. For random band-limited divergence-free fields the ratio  $|\alpha(x)|/\|S(x)\|_2$ , at most 1 pointwise since  $\xi$  is a unit vector, attains 0.9999 at  $s = \frac{1}{4}, \frac{1}{2}, 1, 2$  alike (`deplete_ratio.py`): the stretching rate saturates the trivial bound, uniformly in  $s$ .

*Conditionally it is worth exactly  $\beta/2$  in  $s$ ,* as computed above. The same script shows the mechanism switching on as one drives the direction field towards a constant: the ratio falls to 0.14 once the angular spread of  $\xi$  over the region carrying the vorticity is reduced to 0.02. That regime is the hypothesis of [10], and in the limit it is Proposition 7.1.

So the depletion is the conditional mechanism and nothing besides, and it is already accounted for in Theorem 6.3. What remains open is therefore not its size but whether some *other* structure controls a subcritical quantity in three dimensions — which by the count of the previous subsection is the question whether  $s_c < \frac{5}{2}$ , and at  $s = 0$  is the global regularity of the

Euler equation. We have nothing to offer on it, and prefer to say so in that form rather than as a direction to look in.

**Nothing above uses incompressibility.** Beyond  $\text{div } u = 0$  in the transport term and in Lemma 4.2, no property of the volume-preserving bracket is used. The mechanism that uses more is the geometric depletion of vortex stretching through the direction field  $\xi = w/|w|$  [10]. It is worth saying precisely what that would and would not do here, because the obvious hope is misdirected.

What depletion repairs at  $s = 0$  is a *borderline*. Constantin's representation of the stretching rate  $\alpha = \xi \cdot (\nabla u)_{\text{sym}} \xi$  as a principal value with kernel  $|y|^{-3}$  and a geometric factor vanishing as the directions align is exactly Calderón–Zygmund critical; Hölder continuity of  $\xi$  of order  $\beta$  bounds that factor by  $|y|^\beta$  and renders the kernel integrable. For  $s > 0$  there is no borderline to repair: Lemma 3.1 says the kernel is already  $|y|^{2s-3}$  and already integrable, with no appeal to geometry whatever.

The cancellation itself does transfer, and for a reason that makes its scope clear.

**Lemma 7.3** (alignment cancellation, for every  $s$ ). *Let  $s \geq 0$  and let  $\sigma_{ij,k}^{(s)}(k) = -k_j \varepsilon_{iak} k_a |k|^{-2-2s}$  be the symbol of  $T_s$  from Lemma 2.4, in which  $i$  indexes the component,  $j$  the derivative and  $k$  the argument. For unit vectors  $\xi, \xi'$  put  $D_s(k; \xi', \xi) := \xi_i \sigma_{ij,k}^{(s)}(k) \xi_j \xi'_k$ , the factor which the representation of  $\alpha$  integrates against  $|w|$ . Then*

$$D_s(k; \xi', \xi) = - \frac{(k \cdot \xi) \det(\xi, k, \xi')}{|k|^{2+2s}}.$$

*In particular  $D_s$  is linear in  $\xi'$ , vanishes identically when  $\xi' = \xi$ , and  $|D_s(k; \xi', \xi)| \leq |k|^{-2s} |\xi' - \xi|$ .*

*Proof.* Contract:  $\xi_i \xi_j \xi'_k \sigma_{ij,k}^{(s)} = -(k \cdot \xi) \varepsilon_{iak} \xi_i k_a \xi'_k |k|^{-2-2s}$ , and  $\varepsilon_{iak} \xi_i k_a \xi'_k = \det(\xi, k, \xi')$ . At  $\xi' = \xi$  the determinant has a repeated argument. Linearity in  $\xi'$  is immediate, so  $\det(\xi, k, \xi') = \det(\xi, k, \xi' - \xi)$ , which Hadamard bounds by  $|k| |\xi' - \xi|$ ; combined with  $|k \cdot \xi| \leq |k|$  this gives the estimate. Since the contracted symbol vanishes identically at  $\xi' = \xi$ , so does the contracted kernel.  $\square$

The  $s$ -dependence sits entirely in the scalar  $|k|^{-2-2s}$  and cannot disturb the determinant, which comes from the  $\varepsilon$  tensor in  $w = \text{curl } m$ : the depletion is carried by the curl, not by the metric, and is in that sense indifferent to which inertia operator is used. It is checked against a brute-force contraction in `depletion_check.py`, where the vanishing at alignment is confirmed to machine zero for  $s = 0, \frac{1}{4}, \frac{1}{2}, 1, 2$  against a symbol of size  $O(1)$ , so that the cancellation is genuine and not a vanishing tensor.

With  $\xi$  Hölder of order  $\beta$  we therefore get  $|\xi(x+y) - \xi(x)| \lesssim |y|^\beta$  and a kernel  $|y|^{2s+\beta-3}$  in place of  $|y|^{2s-3}$ : depletion acts exactly as though  $s$  were replaced by  $s + \beta/2$ . The admissible exponent in Theorem 6.3 would fall from  $p > 3/(2s)$  to  $p > 3/(2s + \beta)$ , and the count of the previous subsection would close at  $s + \beta \geq \frac{5}{2}$  rather than  $s \geq \frac{5}{2}$ . Since  $\beta \leq 1$  for a direction field, the best conceivable version reaches  $s \geq \frac{3}{2}$ , and never  $s = 0$ ; that it cannot reach Euler is the consistency one should demand, [10] being a conditional criterion and not a regularity theorem.

And conditional is the point. Hölder regularity of  $\xi$  is a property of the solution, not a consequence of the conservation law, and  $s_c$  is defined by an unconditional statement. So depletion is a candidate for a criterion sharper than Theorem 6.3 — a worthwhile thing, and one which by Lemma 4.2 and  $\|w\|_{L^p} \lesssim \|w\|_{L^\infty}$  would feed straight into it — and not a candidate for displacing  $s_c$ . We know of no mechanism that would do the latter, and record that statement. We make no claims that it is a direction to look in.

**The estimate reverses for  $s < 0$ .** For  $s < 0$  the multiplier  $T_s$  has positive order  $2|s|$ : it loses derivatives rather than gaining them, Lemma 3.1 is false for every  $p$ , and  $\|\nabla u\|_{L^\infty}$  is controlled by no  $L^p$  norm of  $w$  whatever. This is not a defect of the argument but a statement about where the family should break down: the stretching equation (4) is unchanged, while the constitutive law recovering  $u$  is strictly rougher than Euler's own. We regard it as the analytic shape of the expectation that finite-time breakdown, if it occurs in this family, occurs at negative order.

## 8. SCOPE

Everything in Sections 2–6 is proved, granted the quoted commutator estimates of Kato–Ponce [13] and Calderón [9], the Moser product estimate [14], and the kernel asymptotics of [16]. Solutions are those of Definition 5.1, weakly continuous into  $H^r$  by Lemma 5.2, which is what the arguments here require. Strong continuity into  $H^r$  also holds, by the reversibility of (4) and the argument of Bona and Smith [7]; it is omitted because nothing here uses it, not because it is out of reach. Lemma 3.1 is stated on  $\mathbb{T}^3$ , where the kernel is compactly supported and no far field arises; on  $\mathbb{R}^3$  the argument needs an additive low-frequency term and the statement changes accordingly. The case  $s = 0$  is excluded from Theorem 6.3 and is [3]; the case  $s < 0$  is not treated. The criterion is stated for the isotropic multipliers (1) only, and Lemma 3.1 is the only place where isotropy is used — an anisotropic symbol homogeneous of degree  $-2s$  and smooth away from the origin would do as well. Nothing here bears on the global regularity of the Navier–Stokes or Euler equations, and by the count of Section 7 nothing here can.

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The scripts `bkm_check.py`, `kernel_check.py`, `depletion_check.py`, `scaling_check.py`, `dim2_check.py` and `deplete_ratio.py` underlying the computational statements are available from the author.

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